

PARTHO KUMAR PAUL

Ph.D. Candidate, Organic Chemistry
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EDUCATION

Ph.D., Organic Chemistry

Expected Summer 2026

University of California, Irvine

Advisor: Prof. Jenny Y. Yang

Dissertation: Synthesis and Electrochemical Evaluation of Redox-Active Molecules for CO₂ Capture

Fellowships: Eugene Cota-Robles Fellowship (1st & 4th Year) | Dissertation Year Fellowship (Summer 2025)

B.S., Chemical Engineering (Nanotechnology Emphasis), Minor: Chemistry

2020

University of California, Riverside

UC LEADS Scholar (2018)

TEACHING EXPERIENCE

Faculty Intern (BLOOM Program)

Jan 2025 – May 2025

Irvine Valley College, Irvine, CA

- Developed and delivered lectures, laboratory exercises, and assessments for undergraduate organic chemistry courses.
- Designed original instructional content using evidence-based pedagogy, connecting course concepts to real-world applications in electrochemistry and energy science.
- Coordinated curriculum planning with faculty mentor; gained experience with course management systems and learning outcomes assessment.

Teaching Assistant, Chemistry

Aug 2021 – Present

University of California, Irvine

- Led weekly discussion sections and laboratory sessions for General Chemistry and Organic Chemistry across multiple course sections over 4+ years.
- Designed assessments and detailed grading rubrics; provided targeted written feedback to 100+ students per quarter.
- Coordinated with course instructors and fellow TAs to ensure consistent delivery of course materials and equitable grading practices.
- Developed a reputation for making difficult concepts accessible, particularly for students with limited prior chemistry background.

Private Chemistry & STEM Tutor

2017 – Present

Self-Employed, Southern California

- Delivered one-on-one and small group instruction in General Chemistry, Organic Chemistry, and STEM subjects to undergraduate and pre-medical students for 8+ years.
- Adapted teaching strategies to diverse learning needs, backgrounds, and levels of preparation; developed hundreds of original practice problems and worked examples.
- Supported students from underrepresented and first-generation backgrounds in navigating challenging coursework and building confidence in STEM.

RESEARCH EXPERIENCE

DOE SCGSR Fellow / Visiting Research Scientist

Jan 2026 – Jun 2026

Brookhaven National Laboratory, Upton, NY

- Conducting pulse radiolysis experiments in direct collaboration with senior scientists to investigate electron transfer mechanisms in organic molecules.
- Managing independent research program under DOE protocols; coordinating across institutional boundaries between UCI and BNL.

Graduate Student Researcher

Apr 2024 – Present

Yang Lab, University of California, Irvine

- Designed and synthesized 20+ novel redox-active molecules; characterized using NMR, UV-Vis, FTIR, GC-MS, LC-MS, ESI-MS, and electrochemical methods.
- Built Python-based tools to automate spectroscopic and electrochemical data processing; mentored 3+ undergraduate researchers.
- Led 4 concurrent research projects; developed standardized protocols and prepared manuscripts for publication.

Graduate Student Researcher

Aug 2021 – Mar 2024

Pronin Lab, University of California, Irvine

- Investigated cobalt-catalyzed radical–polar crossover reactions; executed multi-step air-sensitive synthetic campaigns.
- Performed isotopic labeling studies (^{18}O) and multi-technique product characterization (NMR, IR, mass spectrometry).
- Presented research at weekly group meetings and advancement to candidacy exam.

PUBLICATIONS

Paul, P. K.; Lin, C. C.; Yang, J. Y. Design, Synthesis, and Studies of Hydroxymethylated Quinone Derivatives for Electrochemical CO_2 Capture. Manuscript in preparation.

Paul, P. K.; Yang, J. Y. Synthesis and Evaluation of Charged Flavin Derivatives for pH-Swing Mediated CO_2 Capture. In preparation.

AWARDS & FELLOWSHIPS

- DOE Office of Science Graduate Student Research (SCGSR) Fellowship — Brookhaven National Laboratory (2026)
- Eugene Cota-Robles Fellowship — UC Irvine (1st & 4th Year)
- School of Physical Sciences Dissertation Year Fellowship — UC Irvine (Summer 2025)
- UC LEADS Scholar — UC Riverside (2018)

SERVICE & LEADERSHIP

Elected Representative, School of Physical Sciences

May 2024 – Present

Associated Graduate Students, UC Irvine

- Represent 500+ graduate students; advocate for TA training improvements, student success initiatives, and equitable academic policies.
- Collect and communicate stakeholder feedback to faculty and administrators across competing institutional priorities.

TECHNICAL SKILLS

Analytical Methods: NMR, UV-Vis, IR/FTIR, GC-MS, LC-MS, ESI-MS, pulse radiolysis, cyclic voltammetry, bulk electrolysis, isotopic labeling

Programming & Data: Python (data analysis, automation), Matplotlib, Origin, Microsoft Excel, ChemDraw, MestReNova, Adobe Illustrator, Canvas LMS

Languages: English (native), Bengali (fluent)

TEACHING AVAILABILITY

Available to teach during day and evening hours.

Courses available to teach: General Chemistry (lecture and laboratory), Organic Chemistry (lecture and laboratory)